Damianne Sarr

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Education

Lone Star College Houston, TX
Electrical Engineering, Transfer Track — GPA: 3.4 *Expected B.S. May 2029*

- **Involvement:** Secretary, STEM Club (Aug. 2025 – May 2026)

Experience

Engineering Intern, EMPOWER Summer Engineering Internship Program May 2026 – Aug. 2026
American Society of Mechanical Engineers (ASME) *Hybrid*

- Architected the electrical design for a rooftop canopy photovoltaic system targeting a 15–25% offset of a campus building's daytime electrical load, serving as sole electrical engineer on a four-person cross-disciplinary team within a 40+ intern cohort.
- Evaluated string versus microinverter architectures for shading mitigation and developed an interconnection approach compliant with NEC 705 and ASCE 7 rooftop wind uplift requirements.
- Modeled three array configurations and selected a canopy-mounted design projected to generate 18% more energy than rooftop mounting while preserving 4 ft maintenance access pathways.
- Authored project charter, feasibility study, and bill of materials, and scheduled a six-phase project lifecycle using Gantt and PERT charts with critical path analysis.
- Presented final design concept and engineering documentation to an audience of practicing CAD engineers, engineering managers, and hiring managers at program close.

Projects

Multi-Sensor Security Monitoring Station | *STM32G474RE, ESP32, C/C++, UART, Fusion 360* Aug. 2026 – Present

- Architecting a dual-microcontroller monitoring station pairing an ESP32 for sensor acquisition and state machine control with an STM32G474RE serving as a dedicated display and indicator node, linked over UART.
- Validated the full data pipeline end to end: ultrasonic distance capture on the ESP32, transmission to the STM32 over a 115200 baud UART link via interrupt-driven receive, and live rendering on an OLED through a custom 5x7 font renderer.
- Interfaced two 1.3" SH1106 OLED displays on independent I2C buses (radar sweep and system status) plus status LEDs, configuring all peripherals through STM32CubeMX.
- Designed a 1k/2k resistive voltage divider to level-shift the 5V HC-SR04 echo signal to a 3.3V logic input, achieving 180 cm detection range without damaging the microcontroller.
- Modeled a two-axis pan-tilt servo gimbal in Fusion 360 delivering 180° pan and 100° tilt coverage, then measured and iterated CAD fit tolerances against printed test coupons until the servo horn press fit held without slippage.

Interactive Campus Navigation System | *ESP32, Embedded Systems, Fusion 360* Aug. 2025 – May 2026

- Coordinated with the Lone Star North Harris President and Computer Engineering faculty on a five-person team to design an interactive campus navigation system for on-campus deployment.
- Spearheaded 3D CAD design and prototyping of a physical campus map covering 15 buildings and 4 designated parking areas using Fusion 360 and 3D printing.
- Programmed ESP32 microcontrollers to illuminate corresponding buildings on button press, delivering real-time visual wayfinding for new students and visitors.
- Refined the design with a multidisciplinary team for usability, scalability, and administrative approval.

Technical Skills

Embedded & Hardware: STM32 (Nucleo F446RE/G474RE), ESP32, Arduino, bare-metal firmware, UART/I2C, sensor integration, circuit prototyping, soldering
Design & Analysis Tools: Fusion 360, KiCad, LTspice, STM32CubeIDE, CubeMX, 3D printing
Languages: C, C++, Python
Certifications: Smart Automation Certified (SACA #79WCXF), Autodesk Fusion Certified (wBEqS-22WK)